

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0295

White Street 138 kV
November 17, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0295. The Interconnection Customer proposes a 536.2 MW natural gas combined cycle facility interconnecting at the White Street 138 kV substation in the Pennsylvania area. The request was received on December 01, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 214 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$69,224,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

ELKINS GENERATING CTR – PRINCETON IL 138 kV: 302.4% loading (165 MVA) following the outage of the WHITE ST – YONGWOOD 138 kV; DFAX = 100.0%.

For the loss of YOUNGWOOD – WHITE ST 138 kV line, the EASTPORT – PRINCETON IL 138 kV loads to 256.9% of its 191 MVA rating. Project DFAX: 100.0%.

WHITE ST – ELKINS GENERATING CTR 138 kV line: 254.2% loading (201 MVA) following the outage of the YOUNGWOOD – WHITE ST SUBSTATION 138 kV line; DFAX = 100.0%.

For the loss of WHITE ST SUBSTATION – YOUNGWOOD 138 kV, the EASTPORT – RED WILLOW 138 kV Line loads to 230.0% of its 209 MVA rating. Project DFAX: 100.0%.

For the loss of WHITE ST – ELKINS GENERATING CENTER 138 kV line, the WHITE ST – YOUNGWOOD 138 kV loads to 218.6% of its 234 MVA rating. Project DFAX: 100.0%.

For the loss of WHITE ST – YOUNGWOOD 138 kV, the IRN BRIDGE – RED WILLOW 138 kV L loads to 208.6% of its 225 MVA rating. Project DFAX: 100.0%.

For the loss of WHITE ST – YOUNGWOOD 138 kV line, the IRON BRIDGE – ELK PLAIN 138 kV loads to 196.3% of its 234 MVA rating. Project DFAX: 100.0%.

For the loss of YOUNGWOOD – WHITE ST 138 kV, the EK PLAIN – TUSHAR 138 kV loads to 185.3% of its 244 MVA rating. Project DFAX: 100.0%.

For the loss of YOUNGWOOD – WOODLAND TAP 345 kV, the TUSHAR – CORNVILLE 138 kV L loads to 155.1% of its 252 MVA rating. Project DFAX: 11.2%.

The YOUNGWOOD 345-138 kV TX is overloaded to 127.6% of its 379 MVA emergency rating for the loss of the ELKINS GENERATING CTR – WHITE ST SUBSTATION 138 kV. The project contributes 100.0% DFAX (536.2 MW) to this overload.

For the loss of ALLEGHENY ENERGY UNITS 3 4 & 5 – SOUTHEAST VICKSBURG SUBSTATION 500 kV L, the WOODLAND TAP SUBSTATION – SEWARD 345 kV Line loads to 124.6% of its 1,079 MVA rating. Project DFAX: 26.8%.

The WOODLAND TAP – YOUNGWOOD 345 kV is overloaded to 107.9% of its 1,801 MVA emergency rating for the loss of the ALLEGHENY ENERGY UNITS 3 4 & 5 – SOUTHEAST VICKSBURG 500 kV. The project contributes 34.4% DFAX (184.6 MW) to this overload.

For the loss of SUNRISE CENTER – MRRET 500 kV, the WAHIAWA – SUNRISE CENTER SUBSTATION 345 kV loads to 104.5% of its 1,462 MVA rating. Project DFAX: 8.4%.

The TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500/345 kV Autotransformer is overloaded to 101.4% of its 1,690 MVA emergency rating for the loss of the HILLTOP 500/345 kV Transformer Bank. The project contributes 6.4% DFAX (34.5 MW) to this overload.

CORONET – FT HUMBUG 500 kV: 100.5% loading (2,439 MVA) following the outage of the LITTLE CHUTE – FT HUMBUG 345 kV line; DFAX = 9.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ELKINS GENERATING CTR – PRINCETON IL 138 kV	Rebuild line with higher-capacity conductor	\$7,789,000	100% (\$7,789,000)
PRINCETON IL – EASTPORT 138 kV	Rebuild line with higher-capacity conductor	\$2,732,000	100% (\$2,732,000)
WHITE ST – ELKINS GENERATING CTR 138 kV	Rebuild line with higher-capacity conductor	\$2,221,000	100% (\$2,221,000)
EASTPORT – RED WILLOW 138 kV Line	Rebuild line with higher-capacity conductor	\$3,132,000	100% (\$3,132,000)
YOUNGWOOD – WHITE STREET 138 kV line	Rebuild line with higher-capacity conductor	\$1,266,000	100% (\$1,266,000)
IRON BRIDGE – RED WILLOW 138 kV	Rebuild line with higher-capacity conductor	\$8,516,000	100% (\$8,516,000)
IRON BRIDGE – ELK PLAIN 138 kV	Rebuild line with higher-capacity conductor	\$2,577,000	100% (\$2,577,000)
TUSHAR SUBSTATION – ELK PLAIN 138 kV	Rebuild line with higher-capacity conductor	\$1,651,000	100% (\$1,651,000)
CORNVILLE – TUSHAR 138 kV	Rebuild line with higher-capacity conductor	\$4,648,000	61% (\$2,832,000)
YOUNGWOOD 345/138 kV XFMR	Replace transformer with higher-rated unit	\$18,872,000	100% (\$18,872,000)
WOODLAND TAP – SEWARD 345 kV line	Rebuild line with higher-capacity conductor	\$41,458,000	18% (\$7,348,000)
WOODLAND TAP – YOUNGWOOD 345 kV	Reconductor line	\$12,415,000	22% (\$2,755,000)
WAHIAWA – SUNRISE CENTER SUBSTATION 345 kV	Reconductor line	\$8,128,000	13% (\$1,082,000)
TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500/345 kV Autotransformer	Replace transformer with higher-rated unit	\$23,218,000	10% (\$2,411,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CORONET – FT HUMBUG 500 kV	Reconductor line	\$51,683,000	8% (\$4,040,000)
	Total allocated to project		\$69,224,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.